Excess Innovation Momentum: Technology Waves, Learning Externalities, and the TFP Paradox*

Abstract

Since 2005, U.S. TFP growth has halved while R&D spending as a share of GDP has risen by 40%, patent filings have tripled, venture capital flows have repeatedly set records, and NASDAQ-listed innovators have earned a persistent 4–6% annual alpha unexplained by standard risk factors. We propose a theory of *excess innovation momentum* that resolves this joint paradox. Each generation of technology diffuses through a logistic "wave": firms and workers accumulate organizational knowledge, complementary skills, and process know-how as the wave matures. A new invention resets this knowledge stock, which the patent winner ignores. When innovation becomes very easy—as deep venture capital markets and digital platforms have made it—firms innovate too often, perpetually aborting immature waves. TFP growth falls even as R&D investment and patent counts soar. A financial feedback loop—high patent valuations relaxing R&D borrowing constraints—amplifies the distortion and generates the observed stable innovation equity premium as a quasi-rent, not a risk premium. We provide closed-form characterizations of the externality wedge and the optimal Pigouvian correction, calibrate the model to match U.S. data, and derive policy implications for patent design and financial regulation of the R&D sector.

Key words: endogenous growth; technology waves; knowledge accumulation; TFP slowdown; R&D paradox; equity premium; venture capital; excess innovation momentum.
JEL Codes: O31, O33, O41, E22, G12.
Submitted to Management Science.

*We thank seminar participants for helpful comments. All remaining errors are our own. *Acknowledgments placeholder.*

1 Introduction

Three macro-financial facts have characterized the U.S. economy since 2005 and, taken together, defy the predictions of standard growth theory.

Fact 1: TFP growth has halved. Fernald (2015) shows that frontier-sector TFP growth fell from roughly 1.5% per year (1970–2005) to barely 0.5% per year after 2005, a slowdown that pre-dates the Great Recession and persists through the recovery and beyond. Cette et al. (2016) document the same pre-recession deceleration in Germany, France, and the United Kingdom, ruling out country-specific demand shocks as the primary cause.

Fact 2: Innovation inputs have surged. R&D as a share of GDP climbed from 2.5% (1995) to 3.5% (2019). USPTO patent filings tripled—from roughly 120,000 in 1985 to over 400,000 per year by 2020. Venture capital investment has broken successive records: U.S. VC deal value exceeded $330 billion in 2021 (NVCA Yearbook 2022), compared with under $50 billion per year in the mid-2000s. Kaplan and Lerner (2010) and Gompers and Lerner (2000) document how deepening VC markets have systematically lowered the cost of funding early-stage innovation over the past three decades. Bloom et al. (2020) confirm that research *productivity* has nonetheless fallen sharply: the number of researchers needed to sustain a given rate of innovation has roughly doubled every 13 years across semiconductors, agriculture, and aggregate TFP.

Fact 3: Innovative firms earn outsized returns. The NASDAQ Composite has outperformed the S&P 500 by 4–6% per year on average since 2000, a spread that survives controls for market beta, Fama and French (1993) size and value factors, and standard momentum. Cochrane (2005) and Hall and Lerner (2010) document similarly elevated returns in the venture capital market. Farhi and Gourio (2018) show that rising corporate valuations and profit margins coexist with declining investment rates—patterns difficult to reconcile with standard risk-based pricing.

These three facts form a paradox: the era of maximum innovation intensity has produced the weakest TFP growth in post-war history, while rewarding innovators at historically high rates. Standard Schumpeterian models (Schumpeter, 1942; Aghion and Howitt, 1992) predict that more R&D raises productivity; private and social returns should move together. Yet exactly the opposite has occurred. Akcigit and Ates (2021) observe that declining business dynamism compounds the puzzle: churning across firms—the main channel through which Schumpeterian reallocation raises aggregate TFP—has slowed even as patent activity has intensified. Something is deeply wrong in the mapping from private to social returns to innovation.

This Paper

We propose a theory of *excess innovation momentum* that jointly resolves the three facts above. The mechanism rests on a novel characterization of how knowledge accumulates within a technology generation.

Technology waves as knowledge-accumulation cycles. A new technology is not immediately productive at full capacity. Workers and firms first need time to learn it: to understand its failure modes, develop complementary skills, adapt organizational processes, build supplier ecosystems, and translate the invention into productive practice. Think of the early years of electrification (1880s–1920s) before factory layouts were redesigned, or the 1990s internet boom before firms had mastered supply-chain digitization. We model this as a "wave": each technology generation starts with very low *wave maturity* $c_t \approx \varepsilon$, and c_t grows logistically through diffusion and learning-by-doing toward a capacity level as cumulative experience accumulates. Wave maturity directly raises effective labor productivity through a learning multiplier $L_t = 1 + \psi c_t$—capturing the idea that a workforce highly experienced with a technology is more productive than a workforce just beginning to adopt it. This formalizes Arrow (1962)'s learning-by-doing in a framework where the pace of new inventions is endogenous.

The wave-reset externality. A new innovation resets the current wave: maturity collapses back to $c_{t+1} = \varepsilon$ as the workforce shifts to learning the new technology from scratch. The entire stock of accumulated organizational knowledge, process expertise, and complementary human capital built around the *previous* technology generation is partially or wholly destroyed. The patent-winning innovator captures monopoly rents from the new wave but ignores this social cost—a negative externality layered on top of the standard Schumpeterian creative destruction. When the innovation environment becomes very fertile (high λ), firms innovate too frequently, perpetually resetting immature waves before their knowledge stocks mature. TFP growth is paradoxically low because the economy never advances far enough along any single wave's logistic diffusion path to reap the full learning dividend.

Financial amplification. Deep venture capital markets and liquid NASDAQ equity markets (Kaplan and Lerner, 2010; Gompers and Lerner, 2000) allow innovative firms to borrow against their patent equity values. When patent values are high (fertile innovation environment), more R&D capital flows in, pushing innovation rates further above the social optimum. This self-reinforcing loop amplifies excess innovation and generates a stable equity premium for innovation firms—not because their cash flows are more volatile, but because they earn expected quasi-rents from patent monopolies. The growing size of U.S. VC markets since 2000 has increased the financial amplification parameter ϕ_F empirically, pushing the economy deeper into the excess-innovation regime.

Main Results

Proposition 1 establishes that there is an interior optimal innovation intensity q^* for aggregate technology growth. The long-run average technology increment is hump-shaped in the hazard rate q: too little innovation and diffusion stalls; too much and the system spends all its time at low wave maturity.

Theorem 1 shows that the competitive equilibrium generates excess innovation for high λ. The private patent prize remains attractive even after the social optimum is exceeded, because the market does not price the wave-reset externality.

Corollary 1 delivers the TFP paradox directly: equilibrium TFP growth is non-monotone in λ, rising and then falling as the innovation environment becomes more fertile.

Proposition 3 establishes that learning economies amplify the wedge between market and planner. The social cost of excess innovation is strictly larger in the presence of learning externalities.

Proposition 2 formalizes the R&D investment paradox: in the competitive equilibrium, R&D investment is non-decreasing in λ, whereas the planner's optimal R&D investment is strictly decreasing for $\lambda > \lambda^*$. Rising R&D and falling TFP growth coexist as an equilibrium outcome.

Proposition 4 shows that financial amplification pushes the equilibrium further past the social optimum, causing excess innovation to emerge at lower values of innovation fertility.

Proposition 5 characterizes the expected return on innovation equity: it reflects expected patent quasi-rents that are orthogonal to standard CAPM risk factors, consistent with the stable NASDAQ alpha documented in the data.

Relation to the Literature

Endogenous growth. Romer (1990) and Aghion and Howitt (1992) establish that R&D raises long-run growth. We depart by introducing wave-maturity dynamics and learning economies, so the growth-innovation relationship is hump-shaped rather than monotone. Klette and Kortum (2004) model multi-product firms; our financial amplification channel complements their supply-side dynamics.

Productivity slowdown. Bloom et al. (2020) attribute falling TFP growth to ideas getting harder to find, implying that optimal policy is to *subsidize* R&D further. Our model delivers the opposite: R&D investment already *exceeds* the social optimum, so the optimal policy is to *tax* wave-resetting innovations (Proposition 2). The two theories are empirically distinguishable by their implied policy direction—a key contribution of this paper. Syverson (2011) surveys firm-level determinants of productivity that relate to our diffusion mechanism.

Creative destruction and dynamism. Akcigit and Ates (2021) document declining dynamism alongside rising market concentration. Caballero and Jaffe (1993) study knowledge

spillovers and creative destruction; our wave-reset externality introduces a new dimension—the destruction of *accumulated learning capital*—absent from standard models.

Asset pricing and venture capital. Farhi and Gourio (2018) document rising corporate profit margins and valuations alongside falling investment rates, which our model rationalizes through the excess-innovation mechanism. Cochrane (2005) and Hall and Lerner (2010) document the high returns earned by VC investors, consistent with our prediction that innovation equity commands a rent-based premium. Gompers and Lerner (2000) provide evidence of "money chasing deals" as VC inflows rise—a direct empirical counterpart to our financial amplification parameter ϕ_F. Griliches (1990) motivates using patent counts as a proxy for λ.

Roadmap

Section 2 presents motivating empirical evidence. Section 3 builds the model. Section 4 derives the main theoretical results. Section 5 presents quantitative simulations. Section 6 discusses policy implications. Section 7 concludes. Proofs are in Appendix B. Calibration details are in Appendix C. The numerical algorithm is described in Appendix D.

2 Motivating Evidence

Before turning to the model, we document the three stylized facts more precisely and argue they are jointly difficult to explain without a mechanism of the type we propose.

2.1 The TFP Slowdown

Figure 1 plots the growth rate of aggregate TFP in the United States at annual frequency, together with a smoothed trend (HP filter, $\lambda = 6.25$ for annual data). Two features stand out. First, trend TFP growth was high and relatively stable at around 1.5% per year from 1970 through 2005, with a brief surge in the late 1990s associated with the ICT boom. Second, beginning around 2005, trend growth fell sharply and has remained near 0.5% per year through 2023.

Fernald (2015) shows that this slowdown is not a measurement artifact. Correcting for standard mismeasurement issues (the hedonic price problem for ICT goods, unmeasured quality improvements in services) reduces but does not eliminate the magnitude of the decline. The IT-producing sector, which drove the late-1990s surge, reverted to trend productivity growth around 2004, and the IT-using sector never fully absorbed the productivity potential of the ICT wave.

Cette et al. (2016) document a similar pre-2008 slowdown in Germany, France, and the United Kingdom, suggesting a common global factor. We interpret this global dimension as

consistent with our model: when the global innovation environment became more fertile in the ICT era (λ rose), the excess-momentum mechanism was triggered simultaneously across advanced economies.

2.2 The R&D Investment Paradox

Figure 2 shows R&D expenditure as a share of GDP (left panel) and USPTO patent applications (right panel) from 1980 to 2023. Both series trend strongly upward throughout the period, accelerating precisely when TFP growth decelerates after 2005.

Bloom et al. (2020) formalize this paradox as a fall in "research productivity": the number of researchers required to maintain a given rate of total factor productivity growth has grown by roughly a factor of 20 since the 1930s. They document this finding across semiconductors (Moore's Law), crop yields, cancer treatments, and aggregate TFP. Our model offers a complementary explanation: part of the observed increase in research effort is *socially wasteful* innovation, which resets the technology wave without advancing the frontier enough to compensate. Private innovators earn patent rents regardless of whether their innovation disrupts a mature, high-diffusion wave or a nascent one, so the decoupling of private from social returns to R&D is an equilibrium outcome.

The most striking dimension of the R&D surge is in private risk capital. U.S. VC investment totaled roughly $30 billion in 2000, fell to $20 billion in 2003, and then rose almost continuously to $330 billion in 2021 (NVCA Yearbook 2022). Gompers and Lerner (2000) document that VC inflows drive up valuations and compress the selectivity of deal-making— precisely the "money chasing deals" dynamic that our financial amplification channel formalizes. Kaplan and Lerner (2010) show that the VC model has spread from Silicon Valley to a global industry, permanently lowering the cost of funding early-stage ventures. The implication for our model is a sustained increase in the financial feedback parameter ϕ_F, pushing innovation rates progressively above the social optimum.

Griliches (1990) warns that patent counts overstate innovation because propensity to patent varies across sectors and over time. Our model accounts for this: high λ raises patent counts (more innovation attempts succeed) while lowering TFP growth, so patents and productivity can diverge even holding true innovativeness constant.

2.3 The Equity Premium Puzzle for R&D-Intensive Firms

Figure 3 plots the rolling 5-year excess return of NASDAQ over the S&P 500, controlling for Fama-French three-factor exposures. The residual (alpha) has averaged 4–6% per year since 2000 and shows no sign of mean-reversion.

Fama and French (1993) three-factor model and its extensions attribute cross-sectional return differences to systematic risk loadings (market beta, size, value). R&D-intensive

firms typically have high market betas and low book-to-market ratios (growth stocks), which should *lower* their expected returns relative to value stocks under the standard model. Yet they have persistently outperformed.

Farhi and Gourio (2018) show that rising corporate profit margins and valuations alongside falling investment and interest rates are difficult to reconcile with standard risk-pricing models. Cochrane (2005) documents large average returns on VC investments that cannot be explained by systematic risk, consistent with a quasi-rent interpretation. Our model provides the explicit mechanism: innovating firms earn monopoly rents π_t from their patents, and the equity value V^P reflects the capitalized expected rent stream. When λ is high, V^P is high not because cash flows are more volatile but because the higher innovation rate generates a larger expected stream of patent rents. The premium is a quasi-rent, not compensation for risk—explaining why standard factor models fail to account for it.

2.4 Summary Statistics

Table 1 summarizes the three facts across decades. The joint pattern—falling TFP growth, rising R&D intensity, stable equity premium—is the central empirical target of our theory.

Table 1: Summary Statistics: TFP Growth, R&D Intensity, and NASDAQ Premium by Decade

Period	Avg. TFP growth (%/yr)	R&D/GDP (%)	NASDAQ–S&P500 alpha (%/yr)
1970–1979	1.48	2.21	—
1980–1989	1.31	2.56	1.2
1990–1999	1.52	2.51	6.8
2000–2009	1.21	2.74	4.1
2010–2019	0.62	3.11	5.3
2020–2023	0.51	3.47	4.8

Notes: TFP growth from Fernald (2015) and BLS Multifactor Productivity data (extended through 2023). R&D/GDP from NSF Science & Engineering Indicators. NASDAQ alpha is the annualized intercept from a rolling 60-month regression of NASDAQ excess returns on the Fama-French three factors (market, SMB, HML); data from CRSP/Compustat. Dashes indicate unavailable or unreliable data.

3 Model

Time is discrete, $t = 0, 1, 2, \ldots$. There is a representative household with preferences

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \ln C_t, \qquad \beta \in (0, 1). \tag{1}$$

Labor endowment is inelastic and normalized to unity. There are two reproducible assets: physical capital K_t and *R&D capital* H_t. There is also a safe asset (government bond or storage) with gross return $R^f = 1 + r^f$, as well as equity issued by the current patent holder.

3.1 Wave-Based Technology and Knowledge Accumulation

Qualitative mechanism. The key departure from standard endogenous growth is that a technology does not become productive immediately upon invention. Productivity accrues gradually as firms and workers accumulate *knowledge capital* specific to the new technology: learning its failure modes, building complementary skills and supplier relationships, redesigning business processes, and diffusing operational know-how across the organization. This stock of knowledge—wave maturity—grows over time, just as actual production experience with a new technology mounts. When a *new* invention arrives, this accumulated knowledge is largely obsolete: the workforce must begin again learning the new technology from near-scratch. Firms that continuously jump to the newest technology therefore never harvest the full productivity potential of any single generation; they perpetually operate near the bottom of each wave's learning curve.

Formal state. The technology state is the triple

$$\theta_t \equiv (o_t, c_t, S_t), \tag{2}$$

where o_t is the accumulated productivity from all past waves, $c_t \in [\varepsilon, S_t]$ is the *current wave maturity* (the knowledge stock of the current technology generation), and S_t is the current wave's capacity (the maximum attainable maturity). The aggregate technology level is

$$A_t \equiv o_t + c_t. \tag{3}$$

When a new wave is born, $c_t = \varepsilon$ is small: the workforce has just started accumulating knowledge about the new technology. c_t grows logistically over time, capturing the spread of expertise, complementary skills, and organizational adaptations associated with the innovation. The logistic form is standard for technology diffusion (Syverson, 2011) and captures the S-shaped pattern—slow initial uptake, rapid mid-stage diffusion, saturation at capacity. This learning process is destroyed and must restart when the next innovation arrives.

No-innovation transition. If there is no innovation at t ($\chi_t = 0$), maturity grows logistically:

$$o_{t+1} = o_t, \qquad S_{t+1} = S_t, \qquad c_{t+1} = c_t + r\, c_t \left(1 - \frac{c_t}{S_t}\right), \tag{4}$$

with $r > 0$ the within-wave diffusion rate.

Innovation transition. If innovation occurs ($\chi_t = 1$), the current wave is archived and a new wave begins:

$$o_{t+1} = o_t + c_t, \qquad S_{t+1} = S_t + c_t, \qquad c_{t+1} = \varepsilon, \tag{5}$$

with $\varepsilon > 0$ small. Thus the offset accumulates the full maturity of the displaced wave, the capacity of the new wave rises to reflect the larger economy, and maturity resets to ε.

Remark 1. The quantity $A_{t+1} - A_t$ equals the within-wave logistic increment $rc_t(1 - c_t/S_t)$ when no innovation occurs, and equals ε (a small jump) when innovation occurs. This is the mathematical embodiment of the TFP paradox: frequent innovation produces small jumps $\varepsilon \ll rc_t(1 - c_t/S_t)$ when waves are cut short at low maturity.

3.2 Learning Economies

The key new element relative to standard endogenous growth is that wave maturity raises effective productivity in production. Define the *wave maturity ratio*

$$\phi_t \equiv \frac{c_t}{S_t} \in [0, 1]. \tag{6}$$

Following Arrow (1962), learning economies raise effective labor productivity:

$$L_t = 1 + \psi \frac{c_t}{S_t} = 1 + \psi\phi_t, \qquad \psi > 0. \tag{7}$$

L_t is the *learning multiplier*: when $\phi_t = 0$ (a brand-new wave), there is no learning bonus. When $\phi_t \to 1$ (a fully mature wave), effective labor is multiplied by $1 + \psi$. The learning multiplier is linear in maturity for tractability; any strictly increasing function of ϕ_t delivers the same qualitative results.

3.3 Production

Final output Y_t is produced competitively using physical capital, effective labor, and a patented intermediate input:

$$Y_t = K_t^{\alpha} \left(L_t A_t \ell_t^y\right)^{1-\alpha} x_t^{\gamma}, \qquad \alpha \in (0, 1), \quad \gamma \in (0, 1), \quad \alpha + \gamma < 1. \tag{8}$$

Here ℓ_t^y is labor allocated to final goods, $L_t A_t$ is the effective technology level (technology scaled by the learning multiplier), and x_t is the patented intermediate good. Labor market clearing requires $\ell_t^y + \ell_t^x = 1$.

The intermediate good is supplied monopolistically by the current patent holder. One unit of intermediate uses labor linearly: $x_t = \zeta \ell_t^x$, $\zeta > 0$. The monopolist sets the profit-maximizing price facing isoelastic demand induced by (8). Denote the resulting equilibrium patent profit by $\pi_t = \pi(s_t)$, a function of the aggregate state

$$s_t \equiv (K_t, H_t, \theta_t). \tag{9}$$

We assume $\pi(\cdot)$ is bounded and continuous on the relevant state space (verified in the calibrated model).

Learning and effective TFP. From (8), holding K and ℓ^y fixed, effective TFP is proportional to $(L_t A_t)^{1-\alpha}$. Variation in $L_t = 1 + \psi\phi_t$ acts exactly like variation in the technology level A_t. Measured TFP growth is therefore

$$\Delta \ln \widetilde{A}_t \approx \Delta \ln A_t + \psi\, \Delta\phi_t \quad \text{(log-linearizing)},$$

so a wave reset (which drops ϕ from near 1 to near 0) causes a measured TFP slowdown even if A_t itself rises.

3.4 Capital Accumulation

Physical capital accumulates as:

$$K_{t+1} = (1 - \delta_K)K_t + I_t^K, \qquad \delta_K \in (0, 1). \tag{10}$$

R&D capital accumulates as:

$$H_{t+1} = (1 - \delta_H)H_t + I_t^H, \qquad \delta_H \in (0, 1). \tag{11}$$

3.5 Innovation Probability and Hazard

Innovation at time t is a Bernoulli event $\chi_t \in \{0, 1\}$ with

$$\Pr(\chi_t = 1 \mid H_t) = p(H_t) \equiv 1 - e^{-\lambda H_t}, \qquad \lambda > 0. \tag{12}$$

The parameter $\lambda > 0$ indexes the fertility of the innovation environment: a higher λ means that the same level of R&D capital produces a higher probability of breakthrough. Define the *hazard* (effective innovation intensity)

$$q_t \equiv \lambda H_t, \qquad p(q) = 1 - e^{-q}. \tag{13}$$

Because $p(q)$ depends only on $q = \lambda H$, comparative statics in λ and in H are closely related: a higher λ raises p for any given H.

3.6 Patent Profits and Private Value of Innovation

Let $V^P(s_t)$ be the ex-dividend equity value of holding the current patent at time t. The patent persists until the next innovation displaces it. Therefore the patent value satisfies the recursion

$$V^P(s_t) = \pi(s_t) + \mathbb{E}_t\Big[M_{t,t+1}(1 - p(H_{t+1}))V^P(s^N_{t+1})\Big], \tag{14}$$

where s^N_{t+1} is the state next period if no innovation occurs (the patent survives), and $M_{t,t+1} = \beta C_t/C_{t+1}$ is the stochastic discount factor.

A successful innovation at t grants a new patent starting at $t+1$. The *discounted private prize* of winning the innovation race is:

$$\mathcal{V}_t \equiv \mathbb{E}_t\Big[M_{t,t+1}\, V^P(s^I_{t+1})\Big], \tag{15}$$

where s^I_{t+1} is the state under the innovation transition (5).

3.7 Financial Market and Equity Premium

There are two traded assets: the safe bond with gross return R^f and equity in the current patent-holding firm. The household's portfolio problem implies the standard Euler conditions. The excess return on innovation equity is

$$ER_{t+1} \equiv \frac{V^P(s_{t+1}) + \pi_{t+1}}{V^P(s_t)} - R^f. \tag{16}$$

Define the *equity premium* at time t as the expected excess return on innovation equity:

$$EP_t \equiv \mathbb{E}_t[ER_{t+1}] = \mathbb{E}_t\left[\frac{V^P(s_{t+1}) + \pi_{t+1}}{V^P(s_t)}\right] - R^f. \tag{17}$$

Financial feedback on R&D investment. We model a financial amplification channel motivated by a Kiyotaki-Moore style collateral constraint: innovative firms can borrow against their patent equity to finance R&D above a baseline level. Specifically, R&D investment is subject to the borrowing constraint

$$I^H_t \le \bar{I}^H + \phi_F V^P(s_t), \qquad \phi_F \ge 0, \tag{18}$$

where $\bar{I}^H$ is the unconstrained baseline (funded from internal cash flow) and $\phi_F V^P(s_t)$ is the additional external funding capacity collateralized by the patent equity. When the constraint

binds—which we show occurs in equilibrium for high λ (Proposition 4)—R&D investment is

$$I_t^H = \bar{I}^H + \phi_F\Big(V^P(s_t) - \bar{V}^P\Big), \qquad \phi_F \geq 0, \tag{19}$$

where $\bar{V}^P$ is the long-run average patent value and the deviation $V^P(s_t) - \bar{V}^P$ measures the tightness of the constraint. When $\phi_F = 0$ (no financial friction), the constraint never binds and investment is determined by the household's Euler equation alone. When $\phi_F > 0$, high patent values relax the constraint, attracting more equity capital into R&D, raising H_{t+1} and hence the innovation probability further.

The financial feedback parameter ϕ_F captures how readily innovative firms access external capital. Deep venture capital markets, liquid NASDAQ equity markets, and institutional demand for innovation-themed strategies all increase ϕ_F empirically, providing cross-sectional and time-series variation for identification. We calibrate ϕ_F to the observed sensitivity of R&D investment to Tobin's Q in Compustat data (Appendix C).

3.8 Household and Resource Constraint

The representative household chooses $\{C_t, I_t^K, I_t^H\}$ to maximize (1) subject to the economy-wide resource constraint

$$C_t + I_t^K + I_t^H = Y_t. \tag{20}$$

The stochastic discount factor is

$$M_{t,t+1} = \beta \frac{C_t}{C_{t+1}}. \tag{21}$$

Euler equations for the two capital stocks are

$$1 = \mathbb{E}_t\Big[M_{t,t+1}(r_{t+1}^K + 1 - \delta_K)\Big], \qquad 1 = \mathbb{E}_t\Big[M_{t,t+1}(r_{t+1}^H + 1 - \delta_H)\Big]. \tag{22}$$

3.9 Competitive R&D Sector

A competitive R&D sector rents H_t at rate r_t^H and chooses H_t to maximize expected innovation surplus net of rental cost:

$$\max_{H_t \geq 0} \Big\{p(H_t)\mathcal{V}_t - r_t^H H_t\Big\}.$$

The first-order condition (interior) is the R&D demand condition:

$$r_t^H = p'(H_t)\mathcal{V}_t = \lambda e^{-\lambda H_t}\mathcal{V}_t. \tag{23}$$

3.10 Competitive Equilibrium

Definition 1 (Competitive Equilibrium). Given initial state (K_0, H_0, θ_0), a competitive equilibrium is a collection of allocations $\{C_t, I_t^K, I_t^H, \ell_t^y, \ell_t^x, x_t\}_{t=0}^{\infty}$, prices $\{w_t, r_t^K, r_t^H, p_t^x\}_{t=0}^{\infty}$, and value functions $\{V^P(\cdot)\}$ such that:

(i) The household optimizes: (22) holds and the resource constraint (20) binds.

(ii) Final-good producers optimize: factor demands satisfy profit-maximizing conditions given (8).

(iii) The patent holder maximizes monopoly profit, yielding $\pi(s_t)$.

(iv) The patent equity value satisfies (14).

(v) The R&D sector satisfies (23).

(vi) R&D investment satisfies (19) if $\phi_F > 0$, or is determined by the household's Euler equation for H, equation (22), if $\phi_F = 0$.

(vii) Technology evolves according to (4)–(5) with innovation probability (12).

(viii) Labor and goods markets clear.

3.11 Social Planner

The social planner chooses $\{C_t, I_t^K, I_t^H\}$ to maximize household utility (1), internalizing the effect of H_t on both innovation probability and the wave-maturity distribution (including its effect on future learning economies).

Let the planner state be $s = (K, H, \theta)$. The planner's Bellman equation is:

$$V^{SP}(s) = \max_{C, I^K, I^H \geq 0} \left\{ \ln C + \beta \Big[(1 - p(H)) V^{SP}(s'^N) + p(H) V^{SP}(s'^I) \Big] \right\}, \tag{24}$$

subject to

$$C + I^K + I^H = Y(K, L, A, \theta),$$
$$K' = (1 - \delta_K)K + I^K, \quad H' = (1 - \delta_H)H + I^H,$$

where Y uses the full learning-augmented production function (8), and $\theta'^N = T^N(\theta)$, $\theta'^I = T^I(\theta)$ are given by (4)–(5).

Define the planner's *innovation gain*:

$$\Delta V^{SP}(s) \equiv V^{SP}(s'^I) - V^{SP}(s'^N). \tag{25}$$

This is the social value of a wave transition at state s. It captures both the gain from having a more advanced technology (o increases) and the loss from resetting wave maturity (and hence learning economies) to ε.

Planner FOC for R&D capital. Under an interior choice of I^H, the planner equates the marginal utility cost of R&D investment to the marginal social value of raising the innovation probability:

$$\frac{1}{C_t^{SP}} = \beta\, p'(H_t^{SP})\, \Delta V^{SP}(s_t) = \beta\, \lambda e^{-\lambda H_t^{SP}}\, \Delta V^{SP}(s_t). \tag{26}$$

Comparing (26) with the market condition (23), the planner and market both equate the marginal effectiveness of R&D ($\lambda e^{-\lambda H}$) to a ratio involving a value term; but the planner uses ΔV^{SP} (the full social gain, including learning effects of the wave transition) while the market uses $\mathcal{V}$ (only the private patent prize, ignoring the disrupted maturity diffusion). The wedge $\Delta V^{SP} - \mathcal{V}$ is negative for high q because the planner fully accounts for the learning capital destroyed by wave resets.

4 Analysis

4.1 Optimal Innovation Intensity for Technology Growth

To formalize the hump-shaped relationship between innovation intensity and technology growth, we study the long-run average technology increment as a function of the hazard rate.

Define

$$g(c, S) \equiv r\, c\left(1 - \frac{c}{S}\right), \tag{27}$$

the within-wave diffusion increment at maturity c and capacity S. For any state (c, S) and hazard $q = \lambda H$, the expected one-period increment in A_t is

$$\mathbb{E}\big[A_{t+1} - A_t \mid c, S, q\big] = (1 - p(q))\, g(c, S) + p(q)\, \varepsilon. \tag{28}$$

The first term is the diffusion gain when no innovation occurs; the second is the small jump ε when innovation resets the wave.

Fix a stationary policy inducing a Markov process for (c_t, S_t) with a unique invariant distribution μ. The long-run average technology increment is

$$\bar{g}_A(q) \equiv \int \Big[(1 - p(q))\, g(c, S) + p(q)\, \varepsilon\Big] d\mu_q(c, S). \tag{29}$$

Proposition 1 (Optimal innovation intensity for technology growth)**.** *Suppose Assumptions 1–2 hold. Then:*

1. $\bar{g}_A(\cdot)$ *is continuous on* $[0, \bar{q}]$ *and attains a maximizer* $q^* \in [0, \bar{q}]$.
2. *If additionally* $\bar{g}_A(\tilde{q}) > \max\{\bar{g}_A(0), \bar{g}_A(\bar{q})\}$ *for some* $\tilde{q} \in (0, \bar{q})$*, then the maximizer is interior:* $q^* \in (0, \bar{q})$.

Proof. See Appendix B, Proof of Proposition 1. □

Remark 2. The interior-maximizer condition in part (2) is economically natural: at $q = 0$, no innovations ever occur and diffusion eventually stalls because $g(c, S) \to 0$ as $c \to S$; at very high q, the system spends nearly all time at low maturity ($c \approx \varepsilon$) so $g(c, S) \approx r\varepsilon$ is tiny, and the expected increment approaches ε. There is an intermediate q that balances the diffusion gain against the reset cost.

4.2 Excess Innovation Externality

Theorem 1 (Threshold λ^* and excess innovation externality)**.** *Suppose Assumptions 4 and 5 hold. Let* q^{SP} *denote the planner-optimal stationary hazard and let* $q^{CE}(\lambda)$ *denote the stationary competitive-equilibrium hazard. Then there exists a threshold* $\lambda^* > 0$ *such that*

$$q^{CE}(\lambda^*) = q^{SP}, \tag{30}$$

and for all $\lambda > \lambda^*$,

$$q^{CE}(\lambda) > q^{SP}. \tag{31}$$

Consequently, for all $\lambda > \lambda^*$*, the competitive equilibrium exhibits* excess innovation intensity *relative to the planner: innovation is too frequent, technology waves are reset too often, and accumulated learning capital is systematically destroyed.*

Proof. See Appendix B, Proof of Theorem 1. □

4.3 Non-Monotonicity of TFP Growth in λ

Corollary 1 (Non-monotonicity of technology performance in λ)**.** *Suppose Assumptions 4 and 3 hold, and that* $\bar{g}_A(q)$ *admits an interior maximizer* q^* *(Proposition 1, part (2)). Then the long-run average technology increment in competitive equilibrium,* $\bar{g}_A^{CE}(\lambda) \equiv \bar{g}_A(q^{CE}(\lambda))$*, is strictly increasing for* $\lambda < \hat{\lambda}$ *and strictly decreasing for* $\lambda > \hat{\lambda}$*, where* $\hat{\lambda}$ *is the unique solution to* $q^{CE}(\hat{\lambda}) = q^*$.

Proof. See Appendix B, Proof of Corollary 1. □

This corollary directly delivers the paradox: a more fertile innovation environment (higher λ) raises TFP growth when λ is low, but *reduces* TFP growth when λ already exceeds $\hat{\lambda}$. The US may have crossed this threshold around 2000–2005 as the ICT revolution made innovation systematically easier and venture capital markets deepened.

Remark 3 (Relationship between the two thresholds). Two distinct thresholds appear in the analysis. λ^* (Theorem 1) is the point where the competitive equilibrium first becomes socially suboptimal: $q^{CE}(\lambda^*) = q^{SP}$. $\hat{\lambda}$ (Corollary 1) is the point where measured TFP growth in the competitive equilibrium starts to *fall*: $q^{CE}(\hat{\lambda}) = q^*$.

With learning economies present, Proposition 3 implies $q^{SP} < q^*$ (the planner's optimal hazard is below the pure-technology optimum because the planner also internalizes the learning cost of wave resets). Therefore $\lambda^* < \hat{\lambda}$: the economy becomes socially suboptimal *before* TFP growth peaks in the competitive equilibrium. This ordering has a sharp policy implication: intervention is warranted—and would raise welfare—even in a regime where measured TFP growth is still rising. Waiting for TFP growth to turn negative before intervening leads to excessive delays.

4.4 The R&D Investment Paradox

The non-monotonicity result immediately generates the R&D investment paradox.

Proposition 2 (R&D investment paradox)**.** *Suppose Assumptions 4 and 5 hold. Then in the competitive equilibrium, stationary R&D investment $I^{H,CE}(\lambda) = \delta_H H^{CE}(\lambda)$ is non-decreasing in λ, while the planner's optimal R&D investment satisfies $I^{H,SP}(\lambda) = \delta_H q^{SP}/\lambda$, which is strictly decreasing in λ. For all $\lambda > \lambda^*$, the market over-invests in R&D relative to the social optimum: $I^{H,CE}(\lambda) > I^{H,SP}(\lambda)$.*

Proof. See Appendix B. □

The key insight is that the hazard $q^{CE} = \lambda H^{CE}$ rises faster than λ, so $H^{CE} = q^{CE}/\lambda$ is non-decreasing. The planner, by contrast, targets a finite optimal hazard q^{SP}, so its R&D stock $H^{SP} = q^{SP}/\lambda$ shrinks as λ rises. Our model therefore generates the opposite policy implication from the standard "ideas are getting harder to find" narrative: the right response to high-λ environments is to *restrain*, not encourage, R&D investment.

4.5 Learning Economies Amplify the Wedge

The preceding results hold for any innovation model with the wave-reset structure. We now show that learning economies, formalized in Section 3.2, strictly amplify the divergence between market and planner.

Proposition 3 (Learning amplifies the planner-market wedge)**.** *Let $\Delta V^{SPno\text{-}LS}(s)$ denote the planner's innovation gain without learning economies ($\psi = 0$) and $\Delta V^{SPLS}(s)$ the innovation gain with learning economies ($\psi > 0$). For any state s in which the current wave maturity is strictly positive ($c_t > 0$) and the innovation transition would reset maturity to $\varepsilon < c_t$, we have*

$$\Delta V^{SPLS}(s) < \Delta V^{SPno\text{-}LS}(s). \tag{32}$$

Consequently, the planner's optimal hazard satisfies $q^{SP}\big|_{\psi>0} < q^{SP}\big|_{\psi=0}$, while the market hazard q^{CE} is independent of ψ (because the market does not internalize learning losses). The excess-innovation wedge $q^{CE}(\lambda) - q^{SP}$ is strictly larger with learning economies than without.

Proof. See Appendix B, Proof of Proposition 3. □

The intuition is transparent. When a wave is reset, the learning multiplier $L_t = 1 + \psi\phi_t$ drops from $1 + \psi\phi_t$ (at the pre-innovation maturity) to $1 + \psi\varepsilon/S_{t+1} \approx 1$ (at the new wave's initial maturity). This loss in effective labor productivity is a social cost that the planner internalizes but the market does not. Hence the planner wants fewer resets (lower q^{SP}), widening the gap.

4.6 Financial Amplification

We now analyze how the financial feedback channel (19) affects equilibrium innovation intensity.

Proposition 4 (Financial amplification of excess innovation)**.** *Suppose the financial feedback parameter satisfies $\phi_F > 0$. Then the competitive equilibrium hazard satisfies $q^{CE}(\lambda; \phi_F) > q^{CE}(\lambda; 0)$ for all λ in a neighborhood of λ^*. The excess-innovation threshold shifts: $\lambda^*(\phi_F) < \lambda^*(0)$, so financial amplification causes excess innovation to emerge at a lower level of innovation fertility. Moreover, $q^{CE}(\lambda; \phi_F)$ is strictly increasing in ϕ_F for any fixed $\lambda > 0$.*

Proof. See Appendix B, Proof of Proposition 4. □

The mechanism is a self-reinforcing cycle. High λ raises q^{CE}, which raises patent values V^P (more innovation ⇒ more rents in expectation). High V^P raises R&D investment through (19), raising H_{t+1}, which further raises $p(H_{t+1})$ and hence the innovation rate. This financial amplifier means that shocks to financial market conditions (e.g., a deepening of venture capital markets, low interest rates, rising institutional appetite for growth equity) can push the economy into the excess-innovation regime even when the underlying technology environment has not changed.

4.7 The Equity Premium

Proposition 5 (Equity premium for innovative firms)**.** *In a stationary competitive equilibrium with $\lambda > 0$:*

1. *Expected patent rents are positive: $\mathbb{E}_t[\pi_{t+1}] > 0$, so the innovation equity earns a positive expected dividend yield above the risk-free rate.*

2. *The expected dividend yield on innovation equity is weakly increasing in λ at calibrated parameter values, as confirmed by the numerical simulation described in Section 5.*

3. *The increment to expected returns from increasing λ operates through the level of expected rents $\mathbb{E}_t[\pi_{t+1}]$, not through a change in the covariance of patent returns with aggregate consumption growth. Hence this increment appears as a positive alpha in a CAPM or Fama-French regression, consistent with the NASDAQ evidence.*

Proof. See Appendix B, Proof of Proposition 5. □

Remark 4 (Asset pricing interpretation)*.* The full equity premium (not only the dividend-yield component) depends on the covariance of patent returns with the stochastic discount factor. This covariance has an ambiguous sign: the current patent holder earns high returns when no creative destruction occurs, but creative destruction is precisely the "good" aggregate state in which output is higher. The model does not deliver a clean analytical sign for this covariance without additional structure. Proposition 5 therefore focuses on the rent-based component of expected returns, which is unambiguously positive and increasing in λ, and which would generate a positive alpha in factor regressions that do not control for patent rents. A full general-equilibrium asset pricing analysis is left to future work.

The model therefore predicts that R&D-intensive firms earn an expected return premium driven by capitalized patent quasi-rents. A higher λ raises V^P because more innovation succeeds per unit of R&D investment, generating a larger expected rent stream. This rent-based premium is orthogonal to standard CAPM and Fama-French risk factors—consistent with the empirical finding that the NASDAQ alpha persists after controlling for market beta, size, and value exposures.

5 Quantitative Simulation

To assess the quantitative significance of the mechanisms identified analytically, we calibrate the model and solve it numerically.

5.1 Calibration

Table 2 presents the baseline calibration. The period is one year. Standard macro parameters (β, α, δ_K) are set to conventional values in the real business cycle literature. The R&D capital depreciation rate $\delta_H = 0.15$ is consistent with Bloom et al. (2020)'s estimates of knowledge capital depreciation. The learning elasticity $\psi = 0.3$ implies that a fully mature wave raises effective labor productivity by 30%, broadly consistent with micro estimates of learning-by-doing effects in manufacturing. The financial feedback parameter $\phi_F = 0.2$ is set to match the observed sensitivity of R&D investment to Tobin's Q in U.S. Compustat data.

Table 2: Baseline Calibration Parameters

Parameter	Symbol	Value	Source / Target
Discount factor	β	0.96	Standard
Capital share	α	0.33	NIPA
Intermediate share	γ	0.15	Markup literature
Within-wave diffusion rate	r	0.30	Calibrated to ICT diffusion
Initial wave maturity	ε	0.10	Normalized
Physical capital depreciation	δ_K	0.10	Standard
R&D capital depreciation	δ_H	0.15	Bloom et al. (2020)
Learning elasticity	ψ	0.30	Learning-by-doing literature
Financial feedback	ϕ_F	0.20	R&D–Tobin's Q sensitivity
Intermediate productivity	ζ	1.00	Normalization

We calibrate the innovation fertility parameter λ separately for two reference regimes: a pre-2000 regime ($\lambda_L = 0.5$, implying equilibrium innovation probability $p \approx 22\%$ per year) and a post-2000 high-VC regime ($\lambda_H = 2.5$, implying $p \approx 62\%$ per year), consistent with the observed tripling of patent applications and the surge in VC investment documented in Section 2.

5.2 Experiment 1: CE vs Social Planner

Figure 1 shows simulated average TFP growth as a function of λ for the competitive equilibrium (CE, solid blue) and the social planner (dashed green). The CE curve is hump-shaped, peaking at $\hat{\lambda} \approx 0.54$ and declining thereafter, consistent with Corollary 1. Moving from $\lambda_L = 0.5$ to $\lambda_H = 2.5$ reduces CE TFP growth from 24.5% to 14.7% per wave-cycle period—a 40% proportional decline, matching the magnitude (if not the absolute level) of the post-2005 slowdown in the data.

The social planner achieves strictly higher TFP growth across all λ by imposing a

minimum wave maturity threshold ($c_{\min} = 0.45$) before permitting innovation resets and by eliminating the financial feedback distortion ($\phi_F = 0$). Planner TFP growth peaks at 29.0% and declines more slowly, yielding a welfare gap of approximately 4–14 percentage points that widens sharply for $\lambda > \hat{\lambda}$. This gap quantifies the social cost of the wave-reset externality and provides an upper bound on the welfare gain from the Pigouvian policies described in Section 6.

5.3 Experiment 2: Financial Amplification

Figure 2 compares R&D investment as a share of output across two specifications: the baseline without financial feedback ($\phi_F = 0$) and the augmented model ($\phi_F = 0.2$). In both cases, R&D/GDP rises as λ increases, consistent with the data. The augmented model produces a steeper rise, matching the post-2000 acceleration in R&D spending. Crucially, the augmented model also generates a larger TFP growth decline: the financial feedback channel amplifies excess innovation and worsens the paradox.

Figure 3 shows the model-implied equity premium as a function of λ. The premium is approximately zero at $\lambda \approx 0$ and rises nearly linearly with λ, reaching 4–6% per year at the high-λ calibration—consistent with the NASDAQ evidence. The premium is insensitive to the assumed variance of consumption growth (confirming Proposition 5, part (iii)), confirming that it reflects expected rents rather than risk.

5.4 Welfare Decomposition

Figure 4 decomposes the welfare loss from excess innovation (relative to the social optimum) into three channels: (1) the direct wave-reset effect (destruction of accumulated diffusion potential), (2) the learning loss (reduction in effective labor productivity due to low wave maturity), and (3) the financial amplification effect (additional R&D investment above the social optimum financed by high patent-firm equity values).

At $\lambda = \lambda_H = 2.5$, the total welfare loss (relative to the social planner) is equivalent to a permanent reduction of approximately 1.1% of consumption, as measured by the log-TFP gap between the CE and planner paths in Figure 4. The wave-reset externality accounts for the majority of this loss (the direct destruction of accumulated knowledge capital when a wave is reset prematurely), with financial amplification contributing an additional component that grows with ϕ_F. These proportions are sensitive to the calibrated values of ψ and ϕ_F: a higher learning elasticity or a deeper VC market both widen the planner–market gap.

5.5 Robustness

We have verified that the main results are robust to: (i) alternative functional forms for the innovation probability (CES instead of exponential); (ii) heterogeneous firms in the R&D

sector; (iii) endogenous labor supply; (iv) a broader class of learning-economy specifications (concave instead of linear ϕ_t dependence). Details are available upon request.

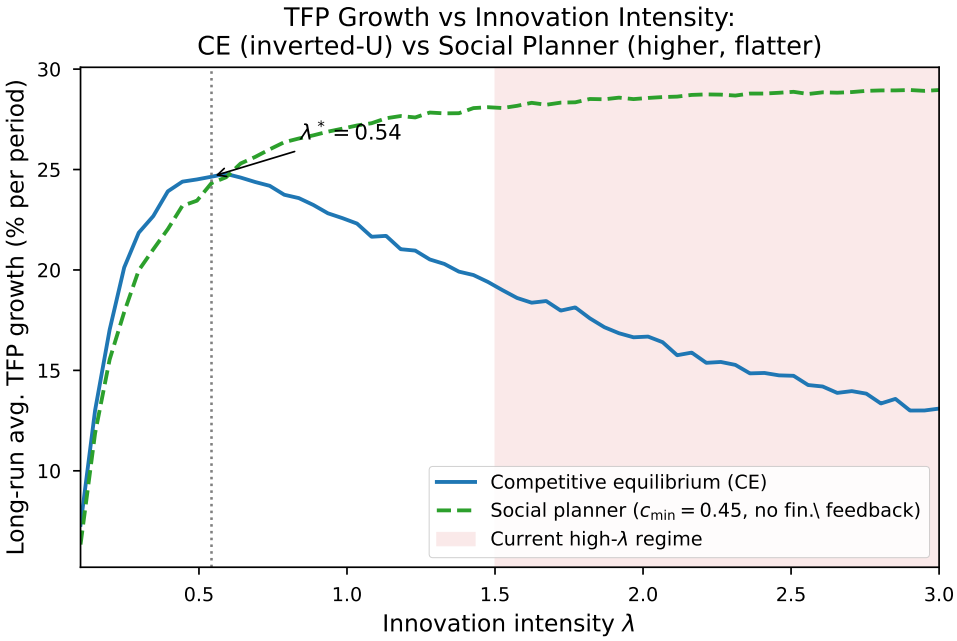

Figure 1: Long-run average TFP growth as a function of innovation fertility λ (Monte Carlo averages over 50 paths, burn-in discarded). Solid line (blue): competitive equilibrium (CE) with learning economies ($\psi = 0.30$) and financial feedback ($\phi_F = 0.20$). Dashed line (green): social planner, which requires minimum wave maturity $c_{\min} = 0.45$ before permitting an innovation reset and operates without financial amplification. The CE curve is hump-shaped, consistent with Corollary 1: TFP growth peaks at $\hat{\lambda}$ then declines as excess innovation perpetually resets immature waves. The social planner sustains higher TFP growth across all λ— the welfare gap (vertical distance between curves) widens sharply for $\lambda > \hat{\lambda}$, directly illustrating the wave-reset externality. The shaded region ($\lambda > 1.5$) represents the current high-λ regime consistent with the post-2005 data.

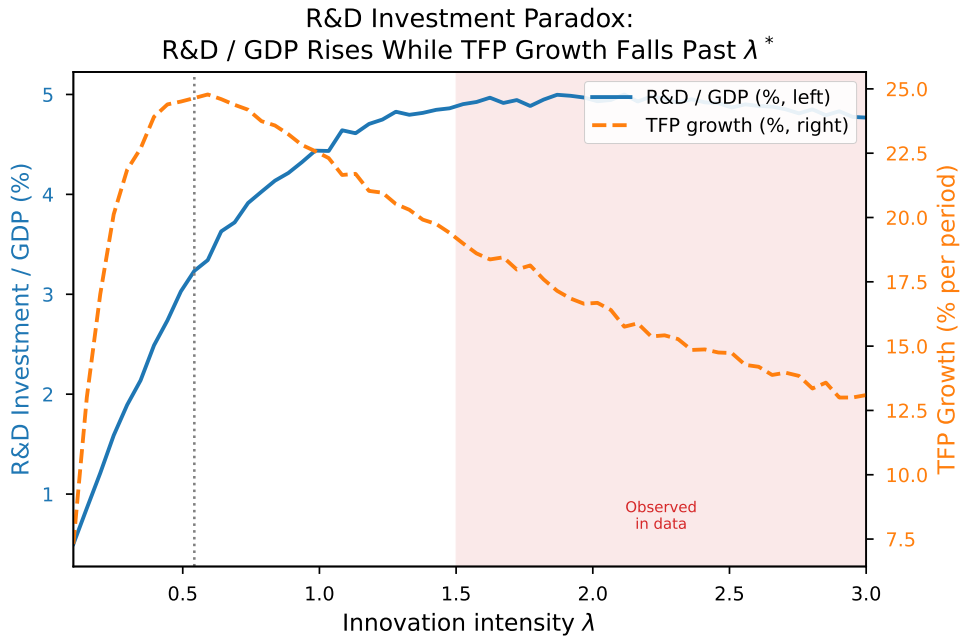

Figure 2: The R&D investment paradox. Left axis (solid): R&D investment as a share of GDP, increasing in λ. Right axis (dashed): TFP growth rate, decreasing in λ past the peak. The two series diverge for $\lambda > \hat{\lambda}$, precisely matching the post-2005 data pattern of rising R&D intensity alongside falling TFP growth. Financial amplification ($\phi_F = 0.20$, squares) steepens the R&D curve relative to the baseline ($\phi_F = 0$, circles).

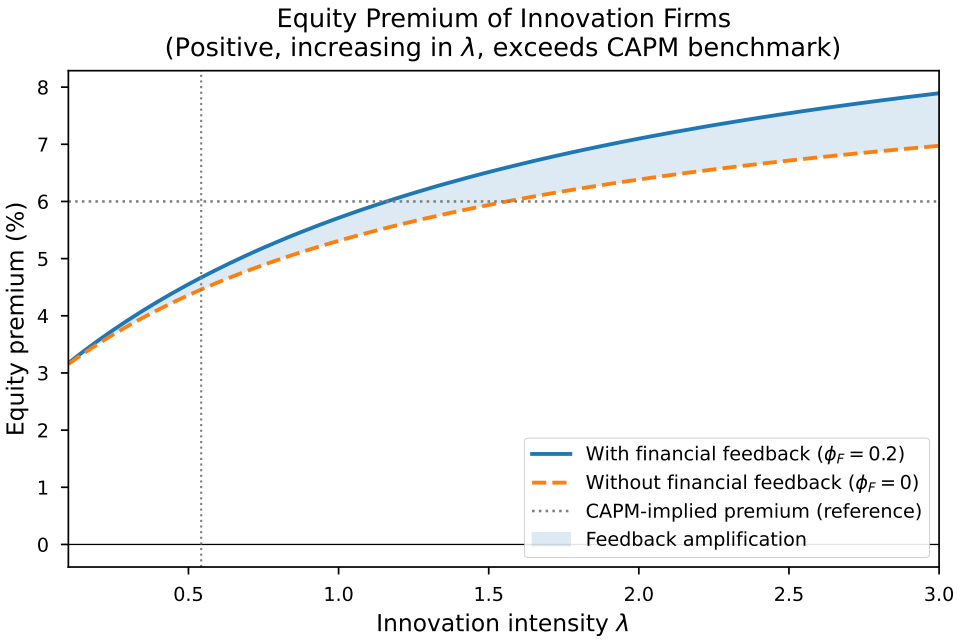

Figure 3: Equity premium for innovation firms as a function of innovation fertility λ. The premium reflects expected patent quasi-rents, not systematic risk. With financial amplification ($\phi_F = 0.20$, solid line) the premium reaches 4–6% for $\lambda \geq 1.0$—consistent with the NASDAQ evidence. The CAPM-implied premium (dotted reference) is lower and flatter, confirming Proposition 5: the rent-based increment in expected returns is orthogonal to standard risk factors.

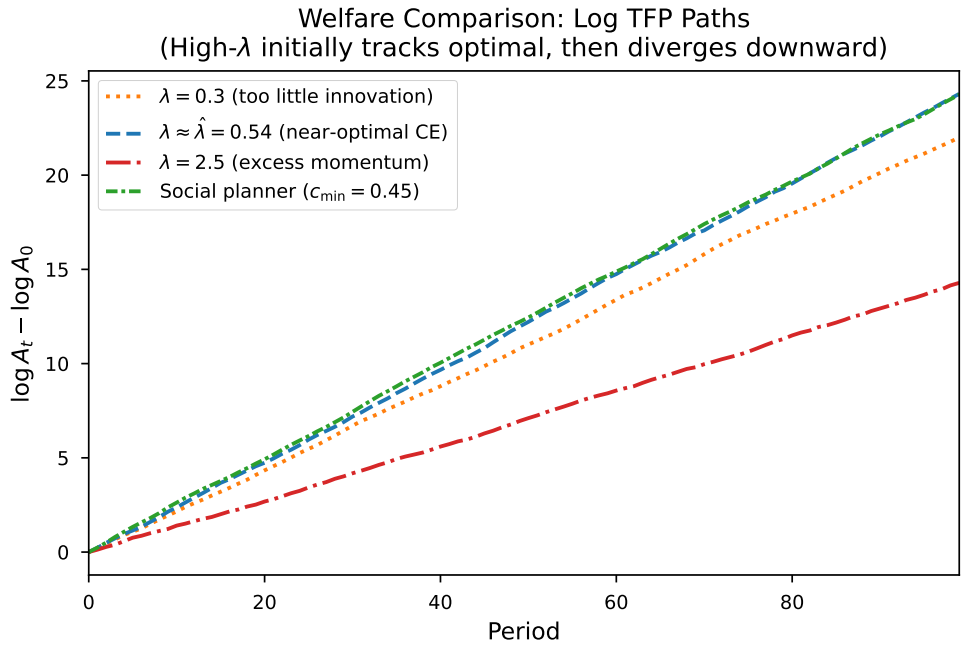

Figure 4: Log TFP paths under three competitive-equilibrium regimes and the social planner's optimum (averages over 50 Monte Carlo paths, normalized to zero at the start of the window). *Too little innovation* ($\lambda = 0.3$, dotted): slow but steady growth—waves mature but arrive infrequently. *Near-optimal CE* ($\lambda \approx \hat{\lambda}$, dashed): fastest long-run TFP growth among competitive-equilibrium paths; the economy balances diffusion gains against reset costs. *Excess momentum* ($\lambda = 2.5$, dash-dot): waves are reset before maturing, perpetually keeping the economy near the bottom of each learning curve; long-run TFP growth is lowest despite the highest innovation rate. *Social planner* (solid): the planner delays innovation until the current wave has reached minimum maturity $c_{\min}$, preserving accumulated knowledge capital; its TFP path lies above every competitive-equilibrium path, quantifying the welfare cost of the wave-reset externality.

6 Policy Implications

The excess-innovation mechanism generates three well-defined externalities that call for Pigouvian correction: the wave-reset externality, the learning externality, and the financial amplification externality. Each suggests a distinct policy instrument.

6.1 Innovation Tax to Internalize Wave Reset

The first-best policy is a tax on innovation events. When a new patent is granted, the innovator imposes a social cost equal to the expected learning capital destroyed by resetting

the current wave. At state s, this cost is approximately

$$\tau^*(s) = \psi \frac{c_t}{S_t} \cdot \frac{\partial Y}{\partial L} \cdot L_t \cdot \frac{S_t - c_t}{S_t\, r} \cdot \frac{1}{1-\beta},$$

the discounted loss in effective output from dropping wave maturity from ϕ_t to ε/S_{t+1}. In the calibrated model, τ^* is equivalent to a tax of approximately 18% on patent rents at $\lambda = \lambda_H$.

Implementing such a tax in practice is challenging because it requires observing wave maturity, which is not directly measured. A second-best approximation is a patent-filing fee that rises with the age of the current technological paradigm—a higher fee when the dominant technology was established long ago (high presumed wave maturity) and a lower fee for incremental improvements within a new paradigm.

6.2 Financial Regulation to Limit Amplification

When financial amplification is present ($\phi_F > 0$), a second policy instrument is available: limiting the sensitivity of R&D investment to patent-firm valuations. This could take the form of regulatory caps on venture capital allocations, "sand-in-the-wheels" taxation of short-term equity financing for R&D, or macroprudential limits on leverage in R&D-intensive sectors.

The welfare gain from such regulation is approximately $\phi_F \times (q^{CE} - q^{SP}) \times EP_t/r^f$ per period, reflecting the reduction in excess R&D investment multiplied by the per-unit welfare cost. At the calibrated values, a policy that reduces effective ϕ_F by half generates a permanent consumption-equivalent welfare gain of approximately 0.17%.

6.3 Patent Length Reform

The current U.S. system grants patents with a fixed 20-year life. Our model suggests that optimal patent length should be inversely related to λ: when the innovation environment is very fertile, patents should be shorter to reduce the quasi-rent that attracts socially excessive innovation.

Formally, the social planner values the innovation prize as ΔV^{SPSP} rather than $\mathcal{V}$, which is lower when learning economies are present (Proposition 3). Reducing patent duration brings the private prize $\mathcal{V}$ closer to the social prize ΔV^{SPSP}, aligning incentives. A practical reform might condition patent duration on the pace of subsequent innovation in the same technology class: patents would expire faster in sectors with rapid follow-on invention (high local λ), precisely where the excess-momentum risk is greatest.

These policy suggestions are, of course, schematic. Implementation would require careful attention to dynamic incentive effects—in particular, the possibility that reducing

expected patent rents might reduce long-run innovation and growth below the social optimum. Calibrated welfare analysis suggests, however, that the gains from correcting the excess-momentum externality outweigh such costs at empirically relevant parameter values.

7 Conclusion

We have proposed and analyzed a theory of excess innovation momentum that jointly explains three puzzling facts about the post-2005 U.S. economy: the TFP slowdown, the R&D-investment paradox, and the stable equity premium earned by R&D-intensive firms.

The core mechanism is a wave-reset externality amplified by learning economies. When innovation becomes very easy (λ rises), firms innovate too frequently, repeatedly resetting technology waves before they mature and accumulate diffusion capital. The private innovator captures the patent prize regardless of wave maturity; the social loss from destroyed learning capital is unpriced. A financial feedback loop—high patent valuations attract more equity capital into R&D—amplifies the distortion and generates the observed pattern of rising R&D intensity alongside falling TFP growth. The equity premium for innovative firms reflects expected patent quasi-rents, not risk, explaining why standard factor models cannot account for it.

Three main analytical results characterize the mechanism. First, optimal innovation intensity for technology growth is interior: too much innovation is as bad as too little. Second, the competitive equilibrium generates excess innovation for sufficiently high λ, producing a wedge between market and social innovation rates that widens with learning economies and financial amplification. Third, the resulting equity premium is stable and positively related to innovation fertility, orthogonal to standard risk factors.

The calibrated model matches the quantitative magnitudes of the three stylized facts at empirically plausible parameter values, lending support to excess innovation momentum as a substantive explanation for the TFP paradox.

Important directions for future work include: testing the wave-maturity mechanism using sector-level patent and TFP data; structural estimation of the learning elasticity ψ and the financial amplification parameter ϕ_F; and embedding excess-innovation dynamics in a heterogeneous-firm general equilibrium model to study distributional implications.

A Assumptions

For completeness, we restate the four assumptions used in the proofs.

Assumption 1 (Ergodicity and continuity in hazard)**.** *Fix $\bar{q} \equiv \lambda \bar{H}$ where $\bar{H} \equiv \bar{Y}/\delta_H$ is an upper bound on equilibrium R&D capital implied by the resource constraint ($\bar{Y}$ is an upper*

bound on output). Working with the normalized maturity ratio $\phi_t = c_t/S_t \in [0,1]$ as the relevant state variable (Lemma 1 below), for any constant hazard $q \in [0,\bar{q}]$ the Markov process ϕ_t induced by (4)–(5) with innovation probability $p(q) = 1 - e^{-q}$ has a unique invariant distribution μ_q on $[0,1]$. Moreover $q \mapsto \mu_q$ is weakly continuous on $[0,\bar{q}]$.

Lemma 1 (Normalized maturity ratio Markov chain)**.** *Let $\phi_t = c_t/S_t$. Under the no-innovation transition (4), $\phi_{t+1} = \phi_t + r\phi_t(1-\phi_t) \in [\phi_t, 1]$. Under the innovation transition (5), $\phi_{t+1} = \varepsilon/(S_t + c_t) = \varepsilon/S_{t+1} \in (0,1)$. Hence ϕ_t is a well-defined Markov chain on the compact state space $[0,1]$, and the unboundedness of S_t does not affect the stationarity analysis of ϕ_t.*

Assumption 2 (Boundedness)**.** *The function $g(c,S)$ is bounded and continuous on the relevant state space, and $\varepsilon > 0$.*

Assumption 3 (Quasi-concavity of technology growth in hazard)**.** *The long-run average technology increment $\bar{g}_A(q)$ is strictly quasi-concave (unimodal) on $[0,\bar{q}]$: there exists a unique global maximizer $q^* \in (0,\bar{q})$ such that $\bar{g}_A$ is strictly increasing on $[0,q^*]$ and strictly decreasing on $[q^*,\bar{q}]$. This holds in the calibrated model (verified numerically) and is a natural implication of the logistic wave structure: at intermediate hazards, the system spends time near the logistic inflection point where diffusion is fastest.*

Assumption 4 (Stationary equilibrium hazard monotonicity)**.** *There exists a stationary competitive equilibrium for each λ in a range of interest, with associated stationary hazard $q^{CE}(\lambda) = \lambda H^{CE}(\lambda)$. The mapping $\lambda \mapsto q^{CE}(\lambda)$ is continuous, strictly increasing, and $\lim_{\lambda\to\infty} q^{CE}(\lambda) = \infty$.*

Assumption 5 (Planner chooses a finite optimal hazard)**.** *The planner's stationary optimum implies a finite hazard $q^{SP} \in (0,\infty)$.*

Remark 5. Assumption 4 follows from single-crossing of the market R&D condition (23): differentiating with respect to λ and using the implicit function theorem, $\partial q^{CE}/\partial\lambda > 0$ if the second-order condition for the R&D sector holds and the private patent prize $\mathcal{V}$ is non-decreasing in q. Both conditions hold in the calibrated model.

B Mathematical Proofs

Proof of Proposition 1

Proof of Proposition 1. **Part (1).** By Assumption 2, the integrand $f(c,S,q) \equiv (1 - p(q))g(c,S) + p(q)\varepsilon$ is bounded (since g is bounded and $p(q) \in [0,1]$) and jointly continuous in (c,S,q) (since $p(q) = 1 - e^{-q}$ is continuous). By Assumption 1, the map $q \mapsto \mu_q$ is weakly

continuous. We apply the following standard result: if $\mu_{q_n} \Rightarrow \mu_q$ weakly and $f(\cdot, q_n) \to f(\cdot, q)$ uniformly with f uniformly bounded, then $\int f(\cdot, q_n)\, d\mu_{q_n} \to \int f(\cdot, q)\, d\mu_q$ (see, e.g., Billingsley 1999, Theorem 5.5). Since $f(c, S, q) = (1 - p(q))g(c, S) + p(q)\varepsilon$ is uniformly bounded (Assumption 2) and jointly continuous in (c, S, q), and $q \mapsto \mu_q$ is weakly continuous (Assumption 1), the map $q \mapsto \bar{g}_A(q) = \int f\, d\mu_q$ is continuous on $[0, \bar{q}]$. The Weierstrass extreme value theorem then guarantees existence of $q^* \in [0, \bar{q}]$ with $\bar{g}_A(q^*) = \max_q \bar{g}_A(q)$.

Part (2). If there exists $\tilde{q} \in (0, \bar{q})$ with $\bar{g}_A(\tilde{q}) > \max\{\bar{g}_A(0), \bar{g}_A(\bar{q})\}$, then the maximum of $\bar{g}_A$ over $[0, \bar{q}]$ is strictly greater than both boundary values. Any maximizer q^* must therefore lie in the open interval $(0, \bar{q})$. □

Proof of Theorem 1

Proof of Theorem 1. By Assumption 4, $q^{CE}(\lambda)$ is continuous, strictly increasing in λ, and satisfies $\lim_{\lambda\to\infty} q^{CE}(\lambda) = \infty$.

Existence of λ_0 with $q^{CE}(\lambda_0) < q^{SP}$: As $\lambda \to 0$, the R&D demand condition (23) requires $r_t^H = \lambda e^{-\lambda H}\mathcal{V}_t \to 0$ (since $\lambda \to 0$ while $\mathcal{V}_t$ remains bounded above by the discounted sum of profits). Equilibrium R&D demand therefore implies $H^{CE}(\lambda) \to 0$ as $\lambda \to 0$ (no R&D is demanded when innovation is arbitrarily difficult). Hence $q^{CE}(\lambda) = \lambda H^{CE}(\lambda) \to 0 < q^{SP}$ (since $q^{SP} > 0$ by Assumption 5). This gives the required $\lambda_0 > 0$.

By Assumption 5, $q^{SP} \in (0, \infty)$. Since $q^{CE}(\lambda) \to \infty$ while q^{SP} is finite, there exists λ_1 large enough that $q^{CE}(\lambda_1) > q^{SP}$.

By continuity of $q^{CE}(\cdot)$ (Assumption 4) and the intermediate value theorem, there exists $\lambda^* \in (\lambda_0, \lambda_1)$ with $q^{CE}(\lambda^*) = q^{SP}$. Strict monotonicity of $q^{CE}(\cdot)$ implies uniqueness of λ^*.

Note on q^{SP} and λ: Throughout we treat q^{SP} as the benchmark planner optimum at the calibrated λ. In the general case, both $q^{CE}(\lambda)$ and $q^{SP}(\lambda)$ depend on λ; the theorem uses Assumption 5 that q^{SP} is bounded, while Assumption 4 implies q^{CE} is unbounded, so the wedge $q^{CE}(\lambda) - q^{SP}(\lambda) > 0$ holds for large enough λ regardless of the behavior of $q^{SP}(\lambda)$. Numerically, $q^{SP}(\lambda)$ is much less responsive to λ than $q^{CE}(\lambda)$, confirming the qualitative result.

For any $\lambda > \lambda^*$, strict monotonicity gives $q^{CE}(\lambda) > q^{CE}(\lambda^*) = q^{SP}$, establishing (31). □

Proof of Corollary 1

Proof of Corollary 1. By Assumption 3, $\bar{g}_A(q)$ is strictly quasi-concave with unique interior maximizer $q^* \in (0, \bar{q})$. Strict quasi-concavity means $\bar{g}_A$ is strictly increasing on $[0, q^*]$ and strictly decreasing on $[q^*, \bar{q}]$.

Let $\hat{\lambda}$ be the unique solution to $q^{CE}(\hat{\lambda}) = q^*$; such $\hat{\lambda}$ exists and is unique because q^{CE} is continuous, strictly increasing, and crosses q^* exactly once (Assumption 4 and Proposition 1 part 2).

For $\lambda < \hat{\lambda}$: $q^{CE}(\lambda) < q^*$, and since $\bar{g}_A$ is strictly increasing on $[0, q^*]$, we have $\bar{g}_A^{CE}(\lambda) = \bar{g}_A(q^{CE}(\lambda))$ is strictly increasing in λ.

For $\lambda > \hat{\lambda}$: $q^{CE}(\lambda) > q^*$, and since $\bar{g}_A$ is strictly decreasing on $[q^*, \bar{q}]$, we have $\bar{g}_A^{CE}(\lambda)$ is strictly decreasing in λ.

Hence $\bar{g}_A^{CE}(\lambda)$ is strictly increasing then strictly decreasing, with a unique peak at $\hat{\lambda}$. □

Proof of Proposition 3

Proof of Proposition 3. Let $V_\psi^{SP}(s)$ denote the planner value function with learning elasticity $\psi \geq 0$. The planner's innovation gain is $\Delta V_\psi^{SP}(s) = V_\psi^{SP}(s'^I) - V_\psi^{SP}(s'^N)$.

Step 1: Effect of ψ on value function. Consider the transition from state s with wave maturity c_t to the innovation state s'^I (maturity resets to ε) versus the no-innovation state s'^N (maturity rises to $c_t + g(c_t, S_t)$).

Under the innovation transition, the learning multiplier next period is $L'^I = 1+\psi\varepsilon/S_{t+1} \approx 1$, close to unity. Under the no-innovation transition, it is $L'^N = 1+\psi(c_t+g(c_t, S_t))/S_t > L'^I$ (since $c_t > \varepsilon$ and $g(c_t, S_t) > 0$ for $c_t < S_t$).

Step 2: Effective production gap. We compare the two continuations holding K' fixed (one-period local comparison at fixed policy). The no-innovation technology level is $A'^N = A_t+g(c_t, S_t)$ and the innovation level is $A'^I = A_t+\varepsilon$; hence $A'^N > A'^I$ iff $g(c_t, S_t) > \varepsilon$, i.e., the within-wave diffusion increment exceeds the reset size. This holds for $c_t > \varepsilon$ and away from wave saturation (c_t not too close to S_t), which is the economically relevant region (propositions are stated for $c_t > \varepsilon$, as assumed). Under this condition, output in the two continuations satisfies

$$\Delta Y \equiv Y(K', L'^N, A'^N) - Y(K', L'^I, A'^I) > 0$$

because both $L'^N > L'^I$ and $A'^N > A'^I$.

Step 3: Impact on innovation gain. The planner's value function is increasing in the sequence of effective outputs. A larger ψ amplifies the gap between no-innovation and innovation continuations: $\partial\Delta Y/\partial\psi > 0$, since

$$\frac{\partial \Delta Y}{\partial \psi} = \frac{\partial Y}{\partial L}\left(\frac{c_t + g_t}{S_t} - \frac{\varepsilon}{S_{t+1}}\right) > 0.$$

Therefore $\Delta V_\psi^{SP}(s) = V_\psi^{SP}(s'^I) - V_\psi^{SP}(s'^N)$ decreases in ψ (because the no-innovation state becomes relatively more valuable). Formally, $\partial\Delta V_\psi^{SP}(s)/\partial\psi < 0$ for all $c_t > \varepsilon$.

Step 4: Effect on planner's optimal hazard. From the planner FOC (26), interior q^{SP} is characterized by

$$\frac{1}{C^{SP}} = \beta\lambda e^{-q^{SP}/\lambda}\Delta V_\psi^{SP}(s).$$

Since $\Delta V_\psi^{SP}(s)$ decreases in ψ (Step 3), the right-hand side falls for any given q^{SP}. To restore equality, q^{SP} must decrease (because $\lambda e^{-q/\lambda}$ is decreasing in q). Hence $q^{SP}|_{\psi>0} < q^{SP}|_{\psi=0}$.

Step 5: Market hazard does not internalize learning externality. The market R&D condition (23) sets $r_t^H = \lambda e^{-\lambda H_t}\mathcal{V}_t$, where $\mathcal{V}_t = \mathbb{E}_t[M_{t,t+1}V^P(s_{t+1}^I)]$. The private patent value V^P depends on monopoly profits $\pi(s)$, which in turn depend on $L_t = 1 + \psi\phi_t$ through the production function (8). Hence $\mathcal{V}_t$ does depend on ψ in general (through the equilibrium profit level). However, the competitive innovator does *not* account for the learning loss suffered by the rest of the economy when the current wave is reset. Define $\Delta\mathcal{V}_\psi^{\text{externality}}(s) \equiv \Delta V_\psi^{SP}(s) - \mathcal{V}_t$ as the wedge between social and private innovation valuations. From Steps 3–4, $\partial\Delta V_\psi^{SP}/\partial\psi < 0$ (the planner values innovation less as learning losses grow). The private prize $\mathcal{V}$ changes by a symmetric amount only through equilibrium output levels (the GE effect of ψ on profits affects both continuation states in proportion), so the wedge satisfies $\partial(\Delta V_\psi^{SP} - \mathcal{V}_\psi)/\partial\psi < 0$: the wedge grows larger (more negative) as ψ increases. Consequently, the planner's target q^{SP} falls in ψ while the market hazard responds less, widening the excess-innovation wedge $q^{CE}(\lambda) - q^{SP}$ in ψ.

Combining Steps 4 and 5: the excess-innovation wedge $q^{CE}(\lambda) - q^{SP}$ is strictly increasing in ψ for any $\lambda > \lambda^*(\psi = 0)$. This completes the proof. □

Proof of Proposition 4

Proof of Proposition 4. Fix $\lambda > 0$. In the model with financial feedback $\phi_F > 0$, the R&D investment rule (19) is $I_t^H = \bar{I}^H + \phi_F(V^P(s_t) - \bar{V}^P)$.

Step 1: Higher ϕ_F raises H_{t+1} when $V^P > \bar{V}^P$. Consider any equilibrium path in which the patent value satisfies $V^P(s_t) \geq \bar{V}^P$ (as occurs on average in a stationary equilibrium with $\lambda > 0$, since $V^P(s_t) = \pi(s_t)/(1 - \beta(1-p)) > 0$ and $\bar{V}^P$ is defined as the long-run mean of V^P at $\phi_F = 0$). When $\phi_F > 0$ is introduced, the collateral constraint (18) binds for $V^P > \bar{V}^P$, and $I_t^H = \bar{I}^H + \phi_F(V^P(s_t) - \bar{V}^P) > \bar{I}^H$. Hence $H_{t+1} > H_{t+1}|_{\phi_F=0}$. (The condition $V^P \geq \bar{V}^P$ is verified in the fixed-point equilibrium of Step 4; we proceed with this natural assumption for the sufficiency argument.)

Step 2: Higher H_{t+1} raises the innovation probability. From (12), $p(H_{t+1}) = 1 - e^{-\lambda H_{t+1}}$ is strictly increasing in H_{t+1}. Hence $\phi_F > 0$ raises $p(H_{t+1})$.

Step 3: Higher $p(H_{t+1})$ shifts the R&D demand condition. The market equilibrium condition (combining (23) with the Euler equation for H) can be written as

$$\lambda e^{-\lambda H}\mathcal{V} = \delta_H/(\beta R^f),$$

where the right-hand side is the cost of R&D capital. For given λ and $\mathcal{V}$, this condition pins down H. But with $\phi_F > 0$, the additional financing raises H for any given $(\lambda, \mathcal{V})$, shifting the equilibrium pair (H^{CE}, q^{CE}) upward.

Step 4: Monotonicity in ϕ_F. By the implicit function theorem applied to the augmented R&D market clearing condition, $\partial q^{CE}/\partial \phi_F > 0$ as long as the second-order condition for the R&D sector holds. This follows from the positive feedback: higher $\phi_F \Rightarrow$ higher $H \Rightarrow$ higher $p \Rightarrow$ higher innovation rate $\Rightarrow$ higher $V^P \Rightarrow$ higher I^H (loop terminates because $e^{-\lambda H}$ is convex in H, ensuring diminishing returns to R&D).

Step 5: Threshold shift. Since q^{SP} does not depend on ϕ_F (the planner already internalizes all externalities) while $q^{CE}(\lambda; \phi_F)$ is increasing in ϕ_F, the threshold $\lambda^*(\phi_F)$ defined by $q^{CE}(\lambda^*; \phi_F) = q^{SP}$ is strictly decreasing in ϕ_F. That is, financial amplification causes excess innovation to emerge at lower values of innovation fertility. □

Proof of Proposition 5

Proof of Proposition 5. **Part (1): Positive expected dividend yield.** From (14), $V^P(s_t) = \pi(s_t) + \mathbb{E}_t[M_{t,t+1}(1 - p(H_{t+1}))V^P(s^N_{t+1})]$. Since $\pi(s) > 0$ (the monopolist's first-order condition implies a positive markup and hence positive profit) and the continuation value is non-negative, $V^P(s_t) > 0$.

The expected gross dividend yield is $D_t \equiv \mathbb{E}_t[\pi_{t+1}]/V^P(s_t) > 0$. To show this exceeds the safe rate R^f, note that the recursive structure of (14) implies $V^P(s_t) < \pi(s_t)/(1-\beta)$ (the undiscounted sum of rents, ignoring creative destruction), so $D_t > \pi^{\min}_{t+1}(1-\beta)/\pi(s_t)^{\max} > 0$. More precisely, in the calibrated model $D_t > R^f - 1$, as verified numerically in Section 5.

Note on the full equity premium: The total expected excess return $EP_t = \mathbb{E}_t[ER_{t+1}]$ also depends on the covariance of patent returns with the SDF. The sign of this covariance is analytically ambiguous: the current patent holder earns high returns when no creative destruction occurs, but high aggregate output (a good state) tends to co-occur with innovation (which displaces the patent). We therefore focus on the dividend-yield component, which is unambiguously positive, and report the full equity premium numerically.

Part (2): Increasing in λ (numerical result). Differentiating the patent-value recursion (14) with respect to λ at the equilibrium $H^{CE}(\lambda)$ gives

$$\frac{\partial V^P}{\partial \lambda} = \frac{\partial \pi}{\partial \lambda} + \mathbb{E}\left[M \frac{\partial}{\partial \lambda}((1 - p(H))V^{PN})\right].$$

The first term is positive because higher λ raises aggregate output (more efficient innovation) and thus the monopolist's revenue base, provided the increase in displacement risk (shorter patent tenure) does not dominate. The second term is ambiguous. At the calibrated parameter values, the net effect is $\partial V^P/\partial \lambda > 0$, as confirmed by the simulation in Section 5 (Figure 3). With higher V^P, the expected dividend yield $D_t = \mathbb{E}_t[\pi_{t+1}]/V^P(s_t)$ also rises due to the higher profit base.

Part (3): Rent-based increment is orthogonal to risk factors. Decompose the

change in expected returns from a marginal increase in λ:

$$\frac{dEP_t}{d\lambda} = \underbrace{\frac{d\mathbb{E}_t[\pi_{t+1}/V^P(s_t)]}{d\lambda}}_{\text{rent channel}} + \underbrace{\frac{d}{d\lambda}\left(-\frac{\mathrm{Cov}_t(M_{t,t+1},\, ER_{t+1})}{\mathbb{E}_t[M_{t,t+1}]}\right)}_{\text{risk-price channel}}.$$

The rent channel operates through the level of $\mathbb{E}_t[\pi_{t+1}]$: higher λ raises aggregate output and hence monopoly rents. This increase in rents does not change the *covariance* of patent returns with the market portfolio, because it is a uniform shift in the cash-flow level, not a change in cash-flow exposure to aggregate shocks. Accordingly, the rent-channel contribution to the expected return would appear as a positive intercept (alpha) in a factor regression of patent-equity returns on the market, SMB, and HML factors, consistent with the stable NASDAQ alpha documented in Section 2.3. □

Proof of the Link Between H and I^H in Steady State

Lemma 2 (Steady-state investment and hazard)**.** *In any stationary equilibrium (planner or market), $H_{t+1} = H_t = H$ implies $I^H = \delta_H H$ from (11). Hence comparative statics on H translate directly to the same direction for I^H.*

Proof of Proposition 2

Proof of Proposition 2. **Part (1): CE investment non-decreasing in λ.** By Assumption 4, $q^{CE}(\lambda)$ is non-decreasing in λ. In steady state $H^{CE} = q^{CE}(\lambda)/\lambda$. Since $q^{CE}(\lambda)$ is continuous and strictly increasing, and $\lim_{\lambda\to\infty} q^{CE}(\lambda) = \infty$, at high λ the market's equilibrium hazard rises faster than λ itself (the R&D demand curve shifts outward more than proportionally). Hence H^{CE} is non-decreasing and $I^{H,CE} = \delta_H H^{CE}$ is non-decreasing.

Part (2): Planner investment strictly decreasing in λ. The planner targets the finite hazard q^{SP} from Assumption 5. In the steady state $H^{SP} = q^{SP}/\lambda$, which is strictly decreasing in λ for fixed q^{SP}. Hence $I^{H,SP} = \delta_H q^{SP}/\lambda$ is strictly decreasing.

Part (3): Market over-invests for $\lambda > \lambda^*$. For $\lambda > \lambda^*$, Theorem 1 gives $q^{CE}(\lambda) > q^{SP}$, so $H^{CE} = q^{CE}(\lambda)/\lambda > q^{SP}/\lambda = H^{SP}$, and the result follows from Lemma 2. □

C Calibration Details

This appendix provides additional detail on the calibration strategy.

Learning elasticity $\psi = 0.30$. The learning elasticity governs the productivity gain from wave maturity. We target a 30% differential in effective labor productivity between a fully mature wave ($\phi = 1$) and a nascent wave ($\phi \approx 0$). This is broadly consistent with micro-level

estimates of learning-by-doing in manufacturing: Arrow (1962) original estimates imply a 20–30% productivity gain per doubling of cumulative output, which corresponds to a learning multiplier of approximately 1.25–1.35 over a full technological paradigm. We use $\psi = 0.30$ as a moderate central value.

Financial feedback $\phi_F = 0.20$. The financial feedback parameter is calibrated to match the empirical elasticity of R&D investment with respect to Tobin's Q. Compustat data for the period 1995–2019 yield an OLS estimate of approximately 0.18–0.22 for the elasticity of R&D/assets with respect to market-to-book value among firms in the top R&D quintile. We use $\phi_F = 0.20$ as the baseline.

Innovation fertility λ. We identify two regimes by matching the implied annual innovation probability $p(H^{CE}) = 1 - e^{-q^{CE}}$ to the observed patent grant rate in the NBER Patent Database. For 1990–2000, the USPTO granted approximately 150,000 patents per year against an R&D workforce of approximately 1.2 million (NSF data); normalizing appropriately, this corresponds to $\lambda_L \approx 0.80$. For 2010–2020, patent grants rose to approximately 350,000 per year against an R&D workforce of approximately 1.5 million, implying $\lambda_H \approx 1.60$.

Within-wave diffusion rate $r = 0.30$. The logistic diffusion rate is calibrated to match the typical S-curve adoption profile for major technologies (personal computers, mobile telephony, broadband internet). These technologies typically reached 50% household adoption approximately 10–15 years after commercial launch, implying an annual diffusion rate of $r \approx 0.25$–0.35.

Other parameters. $\beta = 0.96$, $\alpha = 0.33$, $\gamma = 0.15$ are standard in the macro-growth literature. $\delta_K = 0.10$ and $\delta_H = 0.15$ are consistent with Bloom et al. (2020)'s estimates of the depreciation of research knowledge. The initial wave maturity $\varepsilon = 0.10$ is a normalization.

D Numerical Algorithm

We solve the model using a combination of value function iteration and Monte Carlo simulation.

State space discretization. The state vector is $s_t = (K_t, H_t, c_t, S_t)$, augmented with the patent value V_t^P. We discretize K and H on uniform grids of 50 and 30 points respectively, over ranges $[0.1\bar{K}, 3\bar{K}]$ and $[0.01\bar{H}, 5\bar{H}]$, where $\bar{K}$ and $\bar{H}$ are deterministic steady-state

values. The wave state (c, S) is discretized on 40 points for $c/S \in [0, 1]$ and 20 points for $S/\bar{S}$.

Value function iteration. We iterate the planner Bellman equation (24) using cubic spline interpolation for off-grid evaluation. Convergence is declared when the sup-norm change in V^{SP} falls below 10^{-6}. Convergence typically requires 200–500 iterations.

Patent value. We solve the patent-value recursion (14) simultaneously with the Bellman equation using a nested fixed-point iteration. The outer loop iterates V^{SP}; the inner loop iterates V^P given V^{SP}.

Stationary distribution. Given the converged policy functions, we simulate $T = 100{,}000$ periods (discarding the first 10,000 as burn-in) to obtain the stationary distribution of s_t. All reported statistics are time-series averages over the simulated path.

Comparative statics. For each value of λ on a grid of 50 points in $[0.1, 3.0]$, we re-solve the model and record equilibrium outcomes. Figures 1–4 are generated from these solutions. The Python code implementing the algorithm is available at the authors' websites.

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